

# Research Analysis Report

**Research Topic:** Multi-Agent Systems

**Total Papers:** 5

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## Executive Summary

This report provides a technical overview of 'Multi-Agent Systems' including methodologies, research gaps and experiment planning.

## Literature Review

**Introduction:** This literature review summarizes findings from 5 research papers related to the selected topic. It highlights research trends, methodologies, key contributions, future directions, and overall observations.

### *Paper Summaries*

- {'title': 'Consensus and Cooperation in Networked Multi-Agent Systems', 'year': 2007, 'summary': ''}
- {'title': 'Why Do Multi-Agent LLM Systems Fail?', 'year': 2025, 'summary': 'Despite enthusiasm for Multi-Agent LLM Systems (MAS), their performance gains on popular benchmarks are often minimal. This gap highlights a critical need for a principled understanding of why MAS fail. Addressing this question requires systematic identification and analysis of failure patterns. We introduce MAST-Data, a comprehensive dataset of 1600+ annotated traces collected across 7 popular MAS frameworks. MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems. To enable systematic classification of failures for MAST-Data, we build the first Multi-Agent System Failure Taxonomy (MAST). We develop MAST through rigorous analysis of 150 traces, guided closely by expert human annotators and validated by high inter-annotator agreement ( $\kappa = 0.88$ ). This process identifies 14 unique modes, clustered into 3 categories: (i) system design issues, (ii) inter-agent misalignment, and (iii) task verification. To enable scalable annotation, we develop an LLM-as-a-Judge pipeline with high agreement with human annotations. We leverage MAST and MAST-Data to analyze failure patterns across models (GPT4, Claude 3, Qwen2.5, CodeLlama) and tasks (coding, math, general agent), demonstrating improvement headrooms from better MAS design. Our analysis provides insights revealing that identified failures require more sophisticated solutions, highlighting a clear'}
- {'title': 'Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems', 'year': 2025, 'summary': '"Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive. In this paper, we propose and formulate a new research area: automated failure attribution for LLM multi-agent systems. To support this initiative, we introduce the Who&When; dataset, comprising extensive failure logs from 127 LLM multi-agent systems with fine-grained annotations linking failures to specific agents and decisive error steps. Using the Who&When;, we develop and evaluate three automated failure attribution methods, summarizing their corresponding pros and cons. The best method achieves 53.5% accuracy in identifying failure-responsible agents but only 14.2% in pinpointing failure steps,

with some methods performing below random. Even SOTA reasoning models, such as OpenAI o1 and DeepSeek R1, fail to achieve practical usability. These results highlight the task's complexity and the need for further research in this area. Code and dataset are available at [https://github.com/mingyin1/Agents\\_Failure\\_Attribution](https://github.com/mingyin1/Agents_Failure_Attribution) }

- {'title': 'Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments', 'year': 2017, 'summary': 'We explore deep reinforcement learning methods for multi-agent domains. We begin by analyzing the difficulty of traditional algorithms in the multi-agent case: Q-learning is challenged by an inherent non-stationarity of the environment, while policy gradient suffers from a variance that increases as the number of agents grows. We then present an adaptation of actor-critic methods that considers action policies of other agents and is able to successfully learn policies that require complex multi-agent coordination. Additionally, we introduce a training regimen utilizing an ensemble of policies for each agent that leads to more robust multi-agent policies. We show the strength of our approach compared to existing methods in cooperative as well as competitive scenarios, where agent populations are able to discover various physical and informational coordination strategies.'}

{ 'title': 'Red-Teaming LLM Multi-Agent Systems via Communication Attacks', 'year': 2025, 'summary': 'Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications. While the communication framework is crucial for agent coordination, it also introduces a critical yet unexplored security vulnerability. In this work, we introduce Agent-in-the-Middle (AiTM), a novel attack that exploits the fundamental communication mechanisms in LLM-MAS by intercepting and manipulating inter-agent messages. Unlike existing attacks that compromise individual agents, AiTM demonstrates how an adversary can compromise entire multi-agent systems by only manipulating the messages passing between agents. To enable the attack under the challenges of limited control and role-restricted communication format, we develop an LLM-powered adversarial agent with a reflection mechanism that generates contextually-aware malicious instructions. Our comprehensive evaluation across various frameworks, communication structures, and real-world applications demonstrates that LLM-MAS is vulnerable to communication-based attacks, highlighting the need for robust security measures in multi-agent systems.' }

## Research Trends

- agent
- algorithm
- analyzing
- attribution
- begin
- benchmark
- capability
- case
- challenged
- clues
- collaboration
- complex
- critical
- crucial
- debugging

- deep
- despite
- difficulty
- domain
- enabling
- enthusiasm
- explore
- failure
- failures-provide
- gains
- highlight
- inherent
- language
- large
- learning
- llm-ma
- minimal
- model-based
- multi-agent
- need
- often
- performance
- popular
- principled
- problem-solving
- q-learning
- reinforcement
- remain
- responsible
- revolutionized
- sophisticated
- step
- system
- systems-identifying
- task
- through
- traditional
- underexplored

- understanding

## Research Gaps

- MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems
- These results highlight the task's complexity and the need for further research in this area

## Future Scope

- MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems
- These results highlight the task's complexity and the need for further research in this area

**Conclusion:** Overall, the reviewed studies demonstrate significant progress in this research area while highlighting several opportunities for future work and further experimentation.

## Methodology Comparison

**Total Papers:** 5

### Comparison

- {'title': 'Consensus and Cooperation in Networked Multi-Agent Systems', 'research\_area': 'Agentic AI', 'research\_problem': 'Not Available', 'methodology': ['Not Identified'], 'keywords': [], 'year': 2007, 'citation\_count': 10390, 'quality\_score': 50, 'quality\_classification': 'Good'}
- {'title': 'Why Do Multi-Agent LLM Systems Fail?', 'research\_area': 'Agentic AI', 'research\_problem': 'Despite enthusiasm for Multi-Agent LLM Systems (MAS), their performance gains on popular benchmarks are often minimal', 'methodology': ['Framework', 'Model', 'System', 'Llm', 'Rag'], 'keywords': ['despite', 'enthusiasm', 'multi-agent', 'system', 'performance', 'gains', 'popular', 'benchmark', 'often', 'minimal', 'highlight', 'critical', 'need', 'principled', 'understanding'], 'year': 2025, 'citation\_count': 535, 'quality\_score': 100, 'quality\_classification': 'Excellent'}
- {'title': 'Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems', 'research\_area': 'Agentic AI', 'research\_problem': 'Failure attribution in LLM multi-agent systems-identifying the agent and step responsible for task failures-provides crucial clues for systems debugging but remains underexplored and labor-intensive', 'methodology': ['Model', 'System', 'Llm'], 'keywords': ['failure', 'attribution', 'multi-agent', 'systems-identifying', 'agent', 'step', 'responsible', 'task', 'failures-provide', 'crucial', 'clues', 'system', 'debugging', 'remain', 'underexplored'], 'year': 2025, 'citation\_count': 141, 'quality\_score': 90, 'quality\_classification': 'Excellent'}
- {'title': 'Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments', 'research\_area': 'Agentic AI', 'research\_problem': 'We explore deep reinforcement learning methods for multi-agent domains', 'methodology': ['Approach', 'Algorithm'], 'keywords': ['explore', 'deep', 'reinforcement', 'learning', 'multi-agent', 'domain', 'begin', 'analyzing', 'difficulty', 'traditional', 'algorithm', 'case', 'q-learning', 'challenged', 'inherent'], 'year': 2017, 'citation\_count': 6284, 'quality\_score': 80, 'quality\_classification': 'Very Good'}
- {'title': 'Red-Teaming LLM Multi-Agent Systems via Communication Attacks', 'research\_area': 'Agentic AI', 'research\_problem': 'Large Language Model-based Multi-Agent Systems (LLM-MAS) have revolutionized complex problem-solving capability by enabling sophisticated agent collaboration through message-based communications', 'methodology': ['Framework', 'Model', 'System', 'Llm'], 'keywords': ['large', 'language', 'model-based', 'multi-agent', 'system', 'llm-ma', 'revolutionized', 'complex', 'problem-solving', 'capability', 'enabling', 'sophisticated', 'agent']}

'collaboration', 'through'], 'year': 2025, 'citation\_count': 127, 'quality\_score': 90, 'quality\_classification': 'Excellent'}

### ***Highest Cited Paper***

title: Consensus and Cooperation in Networked Multi-Agent Systems

authors: ['R. Olfati-Saber', 'J. Fax', 'R. Murray']

year: 2007

citation\_count: 10390

summary:

research\_problem: Not Available

methodology: ['Not Identified']

key\_contributions: []

future\_work: []

keywords: []

research\_area: Agentic AI

paper\_score: 50

paper\_quality: Good

### ***Latest Paper***

title: Why Do Multi-Agent LLM Systems Fail?

authors: ['M. Cemri', 'Melissa Z. Pan', 'Shuyi Yang', 'Lakshya A. Agrawal', 'Bhavya Chopra', 'Rishabh Tiwari', 'Kurt Keutzer', 'Aditya G. Parameswaran', 'Dan Klein', 'K. Ramchandran', 'Matei A. Zaharia', 'Joseph E. Gonzalez', 'Ion Stoica']

year: 2025

citation\_count: 535

summary: Despite enthusiasm for Multi-Agent LLM Systems (MAS), their performance gains on popular benchmarks are often minimal. This gap highlights a critical need for a principled understanding of why MAS fail. Addressing this question requires systematic identification and analysis of failure patterns. We introduce MAST-Data, a comprehensive dataset of 1600+ annotated traces collected across 7 popular MAS frameworks. MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems. To enable systematic classification of failures for MAST-Data, we build the first Multi-Agent System Failure Taxonomy (MAST). We develop MAST through rigorous analysis of 150 traces, guided closely by expert human annotators and validated by high inter-annotator agreement ( $\kappa = 0.88$ ). This process identifies 14 unique modes, clustered into 3 categories: (i) system design issues, (ii) inter-agent misalignment, and (iii) task verification. To enable scalable annotation, we develop an LLM-as-a-Judge pipeline with high agreement with human annotations. We leverage MAST and MAST-Data to analyze failure patterns across models (GPT4, Claude 3, Qwen2.5, CodeLlama) and tasks (coding, math, general agent), demonstrating improvement headrooms from better MAS design. Our analysis provides insights revealing that identified failures require more sophisticated solutions, highlighting a clear

research\_problem: Despite enthusiasm for Multi-Agent LLM Systems (MAS), their performance gains on popular benchmarks are often minimal

methodology: ['Framework', 'Model', 'System', 'Llm', 'Rag']

key\_contributions: ['We introduce MAST-Data, a comprehensive dataset of 1600+ annotated traces collected across 7 popular MAS frameworks', 'MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems', 'We develop MAST through rigorous analysis of 150 traces, guided closely by expert human annotators and validated by high inter-annotator agreement ( $\kappa = 0$ )']

future\_work: ['MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems']

keywords: ['despite', 'enthusiasm', 'multi-agent', 'system', 'performance', 'gains', 'popular', 'benchmark', 'often', 'minimal', 'highlight', 'critical', 'need', 'principled', 'understanding']

research\_area: Agentic AI

paper\_score: 100

paper\_quality: Excellent

### ***Common Methodologies***

- Algorithm
- Approach
- Framework
- Llm
- Model
- Not Identified
- Rag
- System

### ***Research Areas***

- Agentic AI

### ***Common Keywords***

- agent
- algorithm
- analyzing
- attribution
- begin
- benchmark
- capability
- case
- challenged
- clues
- collaboration
- complex
- critical
- crucial

- debugging
- deep
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- problem-solving
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- reinforcement
- remain
- responsible
- revolutionized
- sophisticated
- step
- system
- systems-identifying
- task
- through
- traditional

- underexplored
- understanding

### ***Methodology Frequency***

Model: 3  
System: 3  
Llm: 3  
Framework: 2  
Not Identified: 1  
Rag: 1  
Approach: 1  
Algorithm: 1

**Most Common Methodology:** Model

### ***Quality Distribution***

Good: 1  
Excellent: 3  
Very Good: 1

### ***Publication Years***

2007: 1  
2017: 1  
2025: 3

### ***Citation Statistics***

highest: 10390  
lowest: 127  
average: 3495.4  
total: 17477

### ***Comparison Summary***

total\_papers: 5  
most\_common\_methodology: Model  
research\_areas: ['Agentic AI']  
quality\_distribution: {'Good': 1, 'Excellent': 3, 'Very Good': 1}  
citation\_statistics: {'highest': 10390, 'lowest': 127, 'average': 3495.4, 'total': 17477}  
highest\_cited\_title: Consensus and Cooperation in Networked Multi-Agent Systems  
latest\_paper\_title: Why Do Multi-Agent LLM Systems Fail?

# Research Gap Analysis

**Total Papers:** 5

## ***Research Areas***

- Agentic AI

## ***Common Keywords***

- agent
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### ***Future Work***

- MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems
- These results highlight the task's complexity and the need for further research in this area

### ***Research Area Distribution***

Agentic AI: 5

### ***Keyword Frequency***

multi-agent: 4

system: 3

agent: 2

despite: 1

enthusiasm: 1

performance: 1

gains: 1  
popular: 1  
benchmark: 1  
often: 1  
minimal: 1  
highlight: 1  
critical: 1  
need: 1  
principled: 1  
understanding: 1  
failure: 1  
attribution: 1  
systems-identifying: 1  
step: 1  
responsible: 1  
task: 1  
failures-provide: 1  
crucial: 1  
clues: 1  
debugging: 1  
remain: 1  
underexplored: 1  
explore: 1  
deep: 1  
reinforcement: 1  
learning: 1  
domain: 1  
begin: 1  
analyzing: 1  
difficulty: 1  
traditional: 1  
algorithm: 1  
case: 1  
q-learning: 1  
challenged: 1  
inherent: 1  
large: 1  
language: 1

model-based: 1  
llm-ma: 1  
revolutionized: 1  
complex: 1  
problem-solving: 1  
capability: 1  
enabling: 1  
sophisticated: 1  
collaboration: 1  
through: 1

### ***Research Trends***

- multi-agent
- system
- agent
- despite
- enthusiasm
- performance
- gains
- popular
- benchmark
- often
- minimal
- highlight
- critical
- need
- principled

### ***Gap Categories***

datasets: ["MAST-Data is the first multi-agent system dataset to outline the failure dynamics in MAS for guiding the development of better future systems"]

models: []

evaluation: []

applications: []

general: ["These results highlight the task's complexity and the need for further research in this area"]

### ***Emerging Topics***

- multi-agent
- system

- agent

### ***Recommendations***

- Evaluate using standardized benchmarks.

### ***Summary***

dominant\_area: Agentic AI

top\_trend: multi-agent

future\_work\_items: 2

## **Experiment Plan**

### ***Research Areas***

- Agentic AI

### ***Recommended Datasets***

- AgentBench
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets
- OpenAI Evals

### ***Baseline Models***

None

### ***Evaluation Metrics***

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

### ***Hardware Requirements***

cpu: Intel Core i7 / AMD Ryzen 7

ram: 16 GB

gpu: NVIDIA RTX 3060 or better

storage: 100 GB SSD

### ***Validation Strategy***

- Train-Test Split

- K-Fold Cross Validation
- Hyperparameter Tuning
- Statistical Significance Testing

### ***Experimental Workflow***

- Collect Dataset
- Preprocess Data
- Train Baseline Models
- Train Proposed Model
- Evaluate Performance
- Compare Results
- Analyze Errors
- Draw Conclusions

## **Report Summary**

**Dominant Research Area:** Agentic AI

**Top Research Trend:** multi-agent

**Highest Cited Paper:** Consensus and Cooperation in Networked Multi-Agent Systems

**Latest Paper:** Why Do Multi-Agent LLM Systems Fail?

### ***Recommended Datasets***

- AgentBench
- GAIA Benchmark
- HotpotQA
- Hugging Face Datasets
- OpenAI Evals

### ***Baseline Models***

None

### ***Evaluation Metrics***

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

## **Conclusion**

The report has been generated using a personalized multi-agent pipeline. The explanation style was adapted according to the Intermediate profile to improve readability and usefulness.